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ORIGINAL ARTICLE



Stakeholders' Perception on Conservation Outcomes of Forest Protected Area Co-management in Bangladesh

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ABSTRACT

Co-management of forest protected areas (PA) in Bangladesh was initiated in 2004 with the aim of involving forest-dependent people in the PA governance for the conservation of forest resources. This study was conducted in two deliberately selected PA (Lawachara national park and Teknaf wildlife sanctuary) to understand stakeholders' (co-management participants, forest officials, and non-participants of co-management) perception toward conservation outcomes. We interviewed 105 respondents using a pre-tested structured questionnaire, applied generalized logistic regression to model their perception of forest conservation outcomes, and did a content analysis of related literature. Forest officials and non-participants were less likely to perceive that co-management had brought something better for PA in terms of the existence of native tree species (odds ratio (OR) = 0.87), wildlife population (OR = 0.67), and agricultural activities (OR = 0.85) inside the PA, which are in conformity with related published literature. However, co-management participants were found positive (ORs are 1.59, 1.24, and 0.97, respectively) for the above conservation outcomes. The findings of this study would be useful for the Bangladesh forest department to identify the divergent perception of stakeholders and find ways toward better conservation outcomes.

KEYWORDS

Collaborative forest management; biodiversity conservation; forest governance; logistic regression; odds ratio

Introduction

With a population density of 1077 per sq. km (BBS, 2017), pressure on forest resources in Bangladesh is intense. People have been converting forests for settlement, aquaculture, and agricultural purposes (Iftekhar & Hoque, 2005; Masum et al., 2016; M. R. Rahman & Hossain, 2015; Thompson et al., 2017). Thompson et al. (2017) indicated that poverty, over population, lack of land use planning, insufficient capacity to manage and enforce, and unclear land tenure as indirect causes of deforestation in Bangladesh. Government-

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managed natural forests in Bangladesh are historically degraded and deforested due to immense anthropogenic pressures (Iftekhhar & Hoque, 2005; Thompson et al., 2017). These factors pushed the government to change the legal status of government-managed reserve forests into protected areas (PA) although the motive behind PA declaration was failed as it was impractical (Chowdhury et al., 2009; Sharma, 2011) and introduced a conflict between PA managers and local people (Sarker et al., 2017). In this context, policy makers were looking for a participatory and sustainable management strategy. Co-management of forest PA, an arrangement to share power between the forest department and the forest-dependent community (FDC) for forest management (Berkes, 2010, 2009; Carlsson & Berkes, 2005) has been viewed as the best probable solution to the problem.

Co-management has beneath its roots in some developing and underdeveloped countries of Asia and Africa (Arts & De Koning, 2017; Statz et al., 2007; Tanvir & Afroze, 2016). In Bangladesh, co-management has experimentally started its journey with a joint initiative taken by the Bangladesh forest department (BFD) and United States Agency for International Development (USAID) in five pilot PA in 2004. Involving FDC in the governance structure for sustainable management of forest resources as well as to check widespread loss of native biodiversity were the primary objectives of this management approach (NSP, 2006). The Nishorgo Support Project (NSP), the Integrated Protected Area Co-management (IPAC) project, and Climate Resilient Ecosystem and Livelihood (CREL) project supported co-management activities in three different phases in Bangladesh during 2004–2018. At present, there are 54 forest PAs of which 21 are under the system of co-management managed by 28 co-management organizations throughout the country (NSP, 2019).

The outcomes of forest PA co-management in Bangladesh are mixed. Several socio-economic studies tried to capture peoples' perceptions on achievements of forest PA co-management. Alam et al. (2015) conducted a socio-economic study to evaluate the effectiveness of co-management in Teknaf Wildlife Sanctuary (TWS), one of the important co-managed PA operating since 2004. They revealed that due to the adoption of co-management illegal felling had been reduced by 52%, fuel wood collection by 21%, land encroachment for agricultural expansion by 49%, and expansion for settlement by 27%. Another study conducted by Rahman et al. (2017a) in Chunati Wildlife Sanctuary (CWS) assessed the effectiveness of alternative income-generating activities (AIGAs) under the umbrella of co-management showed that the income of project supported group was increased by BDT 2816 and monthly fuel collection was decreased by 257 kg. Mukul et al. (2012) evaluated the peoples' reactions to the outcomes of co-management within just two years of its inception in Satchari National Park, Bangladesh. The study found that forest resources collection decreased gradually and income from non-forestry sources increased sharply because of initiatives taken under co-management activities.

On the other hand, Islam et al. (2020) found an insignificant impact of co-management on tree diversity conservation in important PA of Bangladesh. Meanwhile, widespread non-forestry activities had been reported in some co-managed PA (M. F. Rahman & Islam, 2020; Ullah et al., 2020). R. Rahman et al. (2012) observed weak managerial performance and poor governance system of co-management in TWS and the extraction of resources from the sanctuary was continued as before. Degraded forest areas have increased in CWS during co-management regime (Islam et al., 2018a). With the help of remote sensing and GIS, Islam

et al. (2019a) sketched that co-management failed to halt forest cover loss in major co-managed PA of Bangladesh. Ahmed et al. (2020) revealed the decreasing trend of forest cover in the co-managed forests comparing to conventionally managed forests. In addition, exotic tree species becoming dominant in some co-managed PA of the country (Nath et al., 2016a; Rahman et al., 2017a). Rashid et al. (2013) stated that co-management is still facing some challenges to reach its goals that are hindering conservation efforts.

From the global perspective, the achievements of co-management are also mixed (Arts & De Koning, 2017). Community-based forest management in Ethiopia has been appeared as a successful one in addressing deforestation and livelihood improvement of forest-dependent people (Siraj et al., 2018). Community-based forest conservation was also found successful in Nepal (Thoms, 2008). Turyahabwe et al. (2013) reported a positive impact of collaborative management on forest resources in Uganda. Meanwhile, in Malawi, elected co-management institutions were not empowered to participate actively in the decision-making process (Chinangwa et al., 2016). Rasolofoson et al. (2015) revealed that co-management failed to play a significant role in reducing deforestation in Madagascar. KimDung et al. (2016) suggested devolution of allocative power from centralized government to local administration to ensure effective participation of local people in forest management in Vietnam.

The socioeconomic studies conducted so far regarding forest PA co-management in Bangladesh have been focused on co-management participants' livelihood development. However, the perception of forest officials and non-participants of co-management was somehow omitted. Thus, this study tried to capture the opinion of three principal stakeholders (co-management participants, forest officials and non-participants of co-management) toward conservation outcomes of forest PA co-management. The specific objectives of this study were (1) modeling stakeholders' perception toward the impact of co-management on forest resources conservation, and (2) linking findings of this study with that published research related to outcomes of forest PA co-management in Bangladesh.

Materials and methods

Study sites

Forest PA co-management experimentally started its journey in Bangladesh in 2004 and piloted in Lawachara National Park (LNP), Rema Kelenga Wildlife Sanctuary (RKWS), Satchary National Park (SNP), CWS, and TWS. Later co-management approach was extended in 16 new PA throughout the country in different phases. This study focused on two deliberately selected PA, namely, LNP and TWS from five pilot sites. The LNP is located in the north-eastern and the TWS is situated in the south-eastern part of the country (Figure 1). The LNP was declared as a PA in 1996 and covers an area of 1250 ha. Geographically LNP is located between 24°18'18" N to 24°21'22" N and 91°46'08" E to 91°49'31" E (Figure 1). The landform of the park area is not uniform and it has slopes and hillocks ranging from 10 to 50 m (Islam et al., 2019a). Monthly average temperature ranges from 11.9°C to 33.2°C, average monthly rainfall is 208.83 mm, and humidity ranges from 70% to 85% (BBS, 2011, 2015). LNP houses diverse flora and fauna many of which are rare and endangered. The *Hoolock hoolock* (Hoolock gibbon) is one of the most endangered primates found in LNP (Mittermeier et al., 2006).

91° 18' 00" E (Figure 1). The length of the sanctuary is about 28 km from north-south and the width is 3–5 km from east-west. The sanctuary has a Myanmar boarder on one side and a maiden marine drive way of the country on another side (Islam et al., 2019a). This PA is characterized by a monthly average temperature ranging from 22.12°C (minimum) to 30.3°C (maximum), the monthly average rainfall is 349.40 mm, and monthly average relative humidity ranging from 70% to 85% (BBS, 2015; Khatun et al., 2016). This sanctuary is well known as the habitat of *Elephas maximus* (Asian elephant).

Sampling framework

In this study, targeted samples were co-management participants, non-participants of co-management, and forest officials. Co-management participants are the FDC having involvement in different tiers of co-management governance (Figure 2) and non-participants are forest-dependent local people having no involvement in co-management. Administrative activities of co-management are centered on the forest range office. Co-management governance structure in Bangladesh follows bottom-up approach (Figure 2). Landscape zone falls within 5 km from the PA boundary line (NSP, 2006) and there are several villages within this zone. According to Protected Area Management Rule-2017, a village conservation forum (VCF) is formed with 40–160 members from each village within the landscape zone and two representatives from each VCF form a peoples’ forum (PF). Community patrol group (CPG) consists of 21 members from VCF, and the number of CPG may vary from 3 to 15 based on the area of co-managed PA. Again, representatives from PF, CPG, local government, civil administration, law enforcing agencies, and forest officials form co-management executive committee (CMEC) and co-management general committee (CMGC) in a different formation. CMEC is considered as the most important element of co-management governance structure due to its role and responsibilities. On the other hand, CMGC is merely an advisory body.

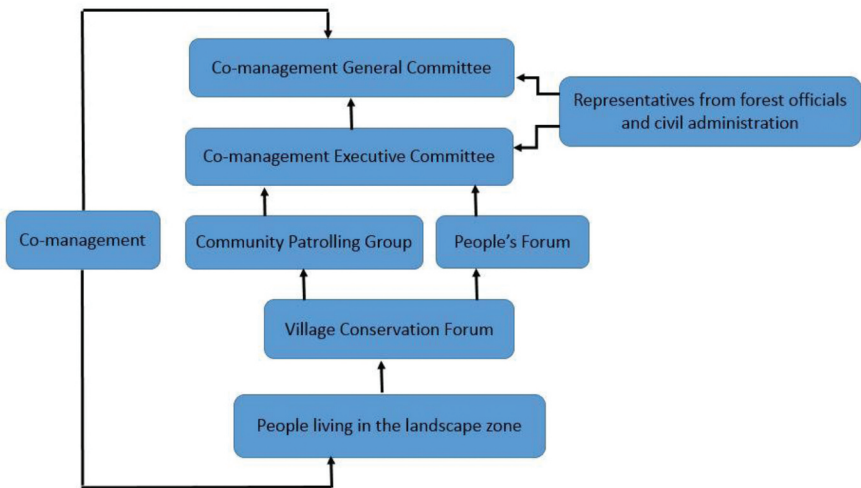


Figure 2. Structure of forest PA co-management governance in Bangladesh.

Table 1. Sampling framework for identification of respondents from the study areas.

PA	Range	Category of respondent	Number of respondent
LNP	Moulvibazar Wildlife Range	Participant of co-management	13
		Non-participant of Co-management	10
		Forest officials	4
TWS	Teknaf Range	Participant of co-management	13
		Non-participant of Co-management	10
	Shilkhali Range	Participant of co-management	13
		Non-participant of Co-management	10
	Whykeong Range	Participant of co-management	13
		Non-participant of Co-management	10
		Forest Officials	9
Total number of respondents			105

We interviewed 52 respondents from co-management participants, 13 from each of the four ranges (Table 1). These 13 participants were randomly selected consisting of three CMEC members, three PF members, three CPG members, and four VCF members. A total of 40 non-participants of co-management (10 from each forest range) living within the landscape zone were interviewed randomly. All forest officials of the selected PA were interviewed. Thus, a total of 105 respondents were interviewed using a semi-structured questionnaire (Table 1).

Quantitative framework

We have used generalized linear model (GLM), popularly known as the logistic regression model to analyze the stakeholders' perceptions toward forest conservation outcomes of co-management activities in selected two PA. Researchers used GLM in socio-economic studies to understand the peoples' perception about biodiversity conservation (Islam et al., 2019a; Jannat et al., 2018; Rahman et al., 2017b). However, GLM are made up of three components: random, systematic, and link function. The random component identifies the response variable ($Y = 0$ or 1) and its probability distribution, the systematic component identifies the set of explanatory variables (X_i), and the link function identifies a linear relationship between the explanatory variables and their probability function. That is,

$$f(P_i(X_i)) = \alpha + \beta_i X_i \quad (1)$$

, where X_i is the vector of socioeconomic attributes of the respondents, $P_i(X_i)$ indicates the probability of respondents' perception about the impact of co-management activities (if attribute increasing = 1, and otherwise = 0)

In GLM, a logit link function can be defined as follows as stated by Islam et al. (2019a):

$$f(P_i(X_i)) = \log\left(\frac{P_i(X_i)}{1 - P_i(X_i)}\right) \quad (2)$$

Combination of equations (1) and (2) gives equation (3).

$$P_i(X_i) = \frac{e^{\alpha + \beta_i X_i}}{1 + e^{\alpha + \beta_i X_i}} \quad (3)$$

Now, if X_i is binary, equation (1) yields the following outcomes:

If and if $X_i = 0$, $f(P_i(0)) = \alpha$

Thus,

$$\beta_i = f(P_i(1)) - f(P_i(0)) = \log \left(\frac{\frac{P_i(1)}{1-P_i(1)}}{\frac{P_i(0)}{1-P_i(0)}} \right) \quad (4)$$

Where odds of $X_i = 1$ and odds of $X_i = 0$ are defined, respectively, as $\frac{P_i(1)}{1-P_i(1)}$ and $\frac{P_i(0)}{1-P_i(0)}$. The ratio of the two is called odds ratio (OR) which represents the odds that an outcome will occur given a particular exposure ($X_i = 1$) compared to the odds of the outcome occurring in the absence ($X_i = 0$) of that exposure.

Thus,

$$OR = \frac{\frac{P_i(1)}{1-P_i(1)}}{\frac{P_i(0)}{1-P_i(0)}} \quad (5)$$

We get equation (6) by combining equation (4) and (5)

$$\beta_i = \log(OR)$$

So,

$$OR = e^{\beta_i} \quad (6)$$

Identification and definition of variables

We interviewed participants of co-management, non-participants of co-management, and forest officials to identify the explanatory variables that might affect their perception of the impact of co-management on forest resources conservation. Through primary screening, the variables with over-lapping information and multicollinearity issues were dropped off in the study. To determine the factors affecting stakeholders' perception toward the impact of co-management, the response variables (Y) used for the logistic regression model were NAT_TREE = Presence of native tree species (1 if native tree species increased and 0 otherwise), WILD_POPU = Wildlife population (1 if wildlife population increased and 0 otherwise), AGRI_INSIDE_PA = Agricultural activities inside PA (1 if agricultural activities inside PA increased and 0 otherwise), and FUEL_WOOD = Fuel wood collection from PA (1 if wildlife population increased and 0 otherwise).

The explanatory variables (X_i) were PA_NAME = Name of PA (1 = LNP, 2 = TWS), GENDER (1 = male, 0 = female), TYPE_RES = Type of respondents (1 = Participants of co-management, 2 = forest officials, and 0 = non-participants of co-management), INCOME = respondents' income in thousand BDT (84.60 BDT = 1 USD), RESIDE = home position (1, 2, 3 if inside, on, and outside the PA boundary, respectively. RESIDE = 1 was assumed as the base case), DURATION = how long the respondent residing in the vicinity of PA in years.

Table 2. List of studies stating outcomes of forest PA co-management in Bangladesh.

Considered parameters	Citation
Forest cover change	Islam et al., 2018a; Islam et al., 2018b; Islam et al., 2019b; Islam et al., 2019a; Ahmed et al., 2020; M. F. Rahman & Islam, 2020; Masum et al., 2020; F. I. Chowdhury et al., 2020
Biodiversity conservation	Nath et al., 2016b; Rahman et al., 2017a; Rahman et al., 2016a; Islam et al., 2020; Masum et al., 2020
Forest governance	Rashid et al., 2013; Rahman et al., 2018; Sarker et al., 2017; R. Rahman et al., 2012; Rashid & Khan, 2014
Forest resources collection	Mukul et al., 2012; Mukul et al., 2014; Rahman et al., 2017c; Alam et al., 2015; Ullah et al., 2020; Nath et al., 2016a
Livelihood improvement	Chowdhury & Koike, 2010; Sohel et al., 2009; Chowdhury et al., 2014; Mukul et al., 2012; Rahman et al., 2017c; Islam et al., 2019b; Tamima, 2016

Content analysis of published articles regarding outcomes of co-management

Several studies (Table 2) have been conducted in different co-managed PA throughout the country. We reviewed 28 published articles and two book chapters (Table 2) to have an insight into how co-management is contributing to forest resources conservation in Bangladesh.

Results

Characteristics of respondents

Table 3 represents that more than 90% of respondents were male. About 50% of respondents were from co-management participants. Of the total respondents (105), 23 (21.90%) from LNP and the rest were from TWS (Table 3). About 45.72% of respondent were replied

Table 3. Characteristics of respondents (N = 105).

Characteristics	n	Percent
Gender		
Male	96	91.43
Female	9	8.57
Category of respondent		
Participant of co-management	52	49.52
Non-participant of Co-management	40	38.10
Forest officials	13	12.38
Respondent from each PA		
LNP	23	21.90
TWS	82	78.10
Monthly income in BDT		
Less than 10000	48	45.72
10000 to less than 30000	39	37.14
30000 to less than 50000	13	12.38
50000 and above	5	4.76
Home position		
Inside the PA boundary	51	48.57
On the PA boundary	22	20.95
Outside the PA boundary	32	30.48
Living duration in years		
Less than 20	28	26.67
20 to less than 40	42	40.00
40 to less than 60	28	26.67
60 and above	7	6.66

that their income level was below BDT 10000 per month whereas, only 4.76% of respondent had monthly income between 50000 and above. About 50% of respondent lived inside the PA boundary and the rest were lived in on the boundary and outside the boundary. The living duration of the most of the respondents in the vicinity of PAs were lies between 20 and 40 years (40%).

Stakeholders' responses on conservation outcomes of forest PA co-management

Table 4 and Table 5 show how the socioeconomic variables affect the stakeholders' perceptions about the impact of co-management on forest conservation outcomes in selected PAs. Of the eight explanatory variables under consideration, PA_NAME2, SEX, TYPE_RES1, TYPE_RES2, RESIDE2, RESIDE3, and DURATION were found significantly associated with the stakeholders' perception toward the influence of forest PA co-management activities on forest resources conservation.

The beta of 'PA_NAME2' was estimated to be -0.178 (Table 4) which was significant at a 5% level, indicates that a respondent from TWS had a smaller $\log(\text{odds})$ of the presence of native tree species compared to a respondent from LNP. The parameter estimate for the variable 'GENDER' was 0.669, which was significant at a 1% level (Table 4). It indicates that the $\log(\text{odds})$ of a male respondent of increasing native tree species due to co-management's initiatives increased by 0.669 compared to that of the same perception of a female

Table 4. Factors that affect the stakeholders' perception about the impact of co-management on biodiversity conservation.

Variable	Perception on presence of native tree species			Perception on wildlife population		
	Estimate(β)	Odds ratio	z value	Estimate(β)	Odds ratio	z value
(Intercepts)	0.094	1.098	0.489	0.861	2.365	4.090**
PA_NAME2	-0.178	0.837	-1.890^*	-0.411	0.663	-3.973^{**}
GENDER	0.669	1.952	4.350**	0.189	1.208	1.118
TYPE_RES1	0.464	1.591	5.259**	0.219	1.244	2.267**
TYPE_RES2	-0.139	0.871	-0.843	-0.402	0.669	-2.221^{**}
INCOME	-0.001	0.999	-0.860	0.001	1.001	1.614
RESIDE2	-0.239	0.787	-2.095^{**}	-0.126	0.882	-0.998
RESIDE3	-0.217	0.804	-2.128^{**}	-0.078	0.925	-0.694
DURATION	-0.007	0.993	-2.564^{**}	-0.009	0.991	-3.103^{**}

Note. ** = significant at 1% level, and * = significant at 5% level.

Table 5. Factors that affect the stakeholders' perception about the impact of co-management on forest dependency.

Variables	Agricultural activities inside the PAs			Fuel wood collection		
	Estimate (β)	Odds ratio	z value	Estimate (β)	Odds ratio	z value
(Intercepts)	0.764	2.146	3.991**	-0.177	0.838	-0.947
PA_NAME2	0.289	1.336	3.082**	0.249	1.283	2.573*
SEX	-0.284	0.752	-1.852^*	0.336	1.399	2.119*
TYPE_RES1	-0.029	0.971	-0.337	-0.221	0.802	-0.321^*
TYPE_RES2	-0.159	0.853	-0.968	0.398	1.489	2.335*
INCOME	0.001	1.001	0.591	0.000	1.000	-1.073
RESIDE2	0.045	1.046	0.392	0.124	1.132	1.049
RESIDE3	0.062	1.064	0.606	-0.028	0.972	-0.273
DURATION	0.001	1.001	0.487	-0.004	0.996	-1.404

Note. ** = significant at 1% level, and * = significant at 5% level.

respondent. The variable 'TYPE_RES1' had the parameter estimate of 0.464 (Table 4), which was significant at 1% level, indicating that the *log(odds)* of a co-management participant of increasing native tree species due to the adoption of co-management increased by 0.464 compared to that of the same perception of the control group (a non-participant of co-management).

Both 'RESIDE2' and 'RESIDE3' had the parameter estimate of -0.239 and -0.217 , respectively, and these were significant at 1% level (Table 4). It indicates that a respondent who lives on and outside the PA boundary line had a smaller *log(odds)* of the presence of native tree species compared to a respondent who lives inside the PA boundary. The parameter estimate for the variable 'DURATION' was -0.007 , which was significant at a 1% level (Table 4). The implication of this finding is that, for having a longer residing period of a respondent, the *log(odds)* of increasing native tree species decreased by 0.007.

The significant beta of 'PA_NAME2' was estimated to be -0.411 (Table 4) which was significant at a 1% level. This finding implies that the *log(odds)* of an individual's (from TWS) perception that the wildlife population is increasing due to the adoption of co-management decreased by 0.411 compared to that of the same perception of an individual from LNP. The beta of 'TYPE_RES1' was estimated to be 0.219 which was significant at a 1% level (Table 4), indicates that a co-management participant had a bigger *log(odds)* of increasing wildlife population compared to a non-participant of co-management. 'TYPE_RES2' had a parameter estimate of -0.402 (significant at 1% level), which indicates that a forest official had a smaller *log(odds)* of increasing wildlife population compared to a non-participant of co-management. The parameter estimate for the variable 'DURATION' was -0.009 , which was significant at a 1% level (Table 4). The implication of this finding is that, for having one year more residing period of a respondent, the *log(odds)* of increasing wildlife population decreased by 0.009.

The beta of PA_NAME2 was estimated to be 0.2896, which was significant at a 1% level (Table 5), indicates that a respondent from TWS had a bigger *log(odds)* of agricultural activities inside the PAs compared to a respondent from LNP. That means the agricultural activities were increasing in TWS. The beta of GENDER was estimated to be -0.2845 , which was significant at a 5% level (Table 5). It indicates that a male respondent had a smaller *log(odds)* of increasing agricultural activities inside the PAs compared to a female respondent. Thus, a male respondent is less likely (Odds is 0.7524, which is less than 1) to perceive that agricultural activity inside the PA was increasing compared to a female respondent.

The parameter estimate of PA_NAME2 was 0.249 which was significant at a 5% level (Table 5), indicates that a respondent from TWS had a bigger *log(odds)* of fuel wood collection compared to a respondent from LNP. That is, the frequency of fuel wood collection was higher in TWS compared to LNP. The parameter estimate for the variable 'GENDER' was 0.336, which was significant at a 5% level (Table 5). It indicates that the *log(odds)* of a male respondent expressing fuel wood collection increasing after the adoption of co-management activities by 0.336 compared to that of the same perception of a female respondent. Thus, a male respondent is more likely (Odds is 1.399, which is greater than 1) to perceive that fuel wood collection is still going on although co-management is working.

The significant beta of TYPE_RES1 was estimated to be -0.221 (Table 5) which was significant at a 5% level. It indicates that a co-management participant had a smaller *log(odds)* of fuel wood collection from co-managed PA compared to a non-participant of co-

management. It means that a person involved in co-management was less likely (Odds is 0.802, which is less than 1.00) to perceive that fuel wood collection was increasing after the adoption of co-management. The beta of TYPE_RES2 was estimated to be 0.398 which was significant at a 5% level (Table 5), indicates that a forest official had a greater *log(odds)* of fuel wood collection compared to a non-participant co-management. This is the indication that a forest official was more likely to perceive that fuel wood collection was increased after the introduction of co-management compared to the same perception of a people having no involvement in co-management.

Linkage between stakeholders’ perception on conservation outcomes and the findings of published articles

A review of selected literature shows both positive and negative impacts of forest PA co-management in Bangladesh (Figure 3). In terms of biodiversity conservation and forest governance, co-management could do very little as researchers reported very low positive outcomes (Figure 3). In addition, out of eight studies regarding forest cover of co-managed PA, five studies revealed that forest cover was in decreasing trend (Figure 3). On the other hand, all studies that evaluated the livelihood improvement of FDC concluded positive outcomes. Findings regarding forest resources collection showed that 50% of articles reported reduction and other 50% of studies found an unchanged pattern of forest resources collection (Figure 3).

The perception of co-management stakeholders can be linked with the findings of published articles related to forest PA co-management outcomes in Bangladesh. Co-management participants were inclined toward the achievement of co-management which might be due to the upliftment of their social status and lifestyle through livelihood improvement projects under the umbrella of co-management. Similar results were also

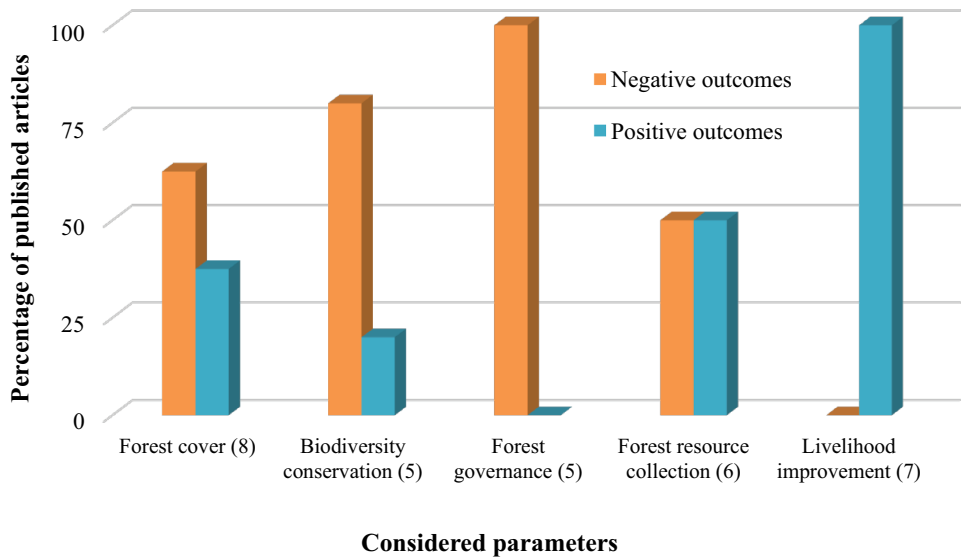


Figure 3. Research outcomes of forest PA co-management in Bangladesh.

reported in published research papers (Figure 3). The perception of forest officials and non-participants of co-management on biodiversity conservation was similar to the reported findings (Figure 3). Forest officials and non-participants of co-management were less likely to perceive the role of co-management on the conservation of native flora and fauna. Agricultural practices inside the forests and the collection of forest resources were negatively correlated with forest cover. Forest officials and non-participants perceived that co-management failed to reduce agricultural activities inside the PA and to halt fuelwood collection from PA. Thus, it led to forest cover loss which was also reported by most of the published articles (Figure 3). Forest cover loss, fuelwood collection, the poor state of floral and faunal diversity, agricultural practices inside the PA are the sign of poor forest governance reported in all articles (Figure 3).

Discussion

Co-management in Bangladesh still struggling to continue its activities for sustainable management of forest resources (R. Rahman et al., 2012; Rashid et al., 2013). The findings of this study show mixed outcomes of forest PA co-management of selected PA. Perception of co-management participants was inclined toward the achievements of co-management. Co-management participants were financially and logistically supported by the foreign donor agencies aided projects (Rahman et al., 2016b) and this might be a probable reason for their inclination toward achievements of co-management. While non-participants' and forest officials' perception was somehow different with regards to forest conservation which is reflected in published articles too (see Figure 3).

Respondents in LNP were more positive toward the presence of native tree species and wildlife in PA compared to the respondents of TWS. Researchers also reported that LNP has diverse native tree species and wild fauna (Aziz, 2011; Uddin & Hassan, 2010). In contrast, TWS has a large geographical area, but fragmented forest patches (Islam et al., 2019a) and the biodiversity of TWS is under continuous threat due to various anthropogenic pressures (Alam et al., 2015; Ullah et al., 2020). Thus, non-participants' perception regarding the presence of native tree species and wild fauna was justified. Respondents who live on and outside the PA boundary line were less likely to perceive that co-management was helping to increase the native tree species in the PA compared to the respondents who live inside the boundary. We suspect that the respondents who live inside the PA boundary were afraid of an eviction and so they responded positively about the status of the forest. Residents with a longer living period in and around of PA were less likely to perceive that co-management was helping increase the native tree species and wildlife in the PA. Older people experienced how a forest looked like in the past days and they can easily trace the improvement or deterioration of a forest with their historical knowledge and observations.

Female respondents were found vocal to express the actual scenario of conservation outcomes of co-management whereas, male respondents were seen as a little loyal toward focusing the success of co-management efforts. Male respondents might think that there would be a chance of getting benefits in foreseeable future from co-management projects if they responded in favor of this management approach. Agricultural activities were in increasing trend in TWS compare to LNP. A similar observation was also made by Alam et al. (2015). They reported that 1000 ha of land in TWS had been encroached for betel leaf cultivation and 2000 households were directly depending on betel leaf cultivation for their

livelihood. In addition, fuelwood collection was comparatively higher in TWS than in the LNP. Ullah et al. (2020) revealed that 1.32 tons of wooden poles were collected from TWS during 2015 to 2017 for betel leaf cultivation. This is also an indication of forest cover loss which is supported by a couple of previous studies (Ahmed et al., 2020; Islam et al., 2018a, 2018b, 2019a).

Conclusion and policy implications

Co-management approach is being widely promoted worldwide for the management of natural resources. Literature reported that outcomes of co-management of forest resources in Bangladesh are mixed and in some cases, we have poor outcomes. BFD needs to work continuously to find out the loopholes behind the poor outcomes of this local-level forest management approach. Since female members were found more straightforward and open-voiced against any irregularity and corruption compared to male members, it is also recommended to help facilitate more female participation in the different tiers of co-management governance. Forest resources collection and agricultural practice inside the PA might be reduced through creating alternative income-generating sources for FDC.

An initiative may be taken under the umbrella of co-management for the conservation of endangered native tree species and wildlife. Native tree species provide both habitat and food for wildlife. Thus, conserving native tree species may promote the conservation of endangered wild fauna. BFD can implement such conservation initiatives by incorporating these issues in forest policy. Encroachment of forest land and collection of fuelwood may be stopped through active participation of all stakeholders of co-management governance. The highest level of dedication of every stakeholder can lead the co-management to a successful journey.

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